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EDUCATION

Expected Jan 2027 **SABANCI UNIVERSITY** – ***B.S. in Computer Science, Faculty of Engineering***
GPA: 3.85/4.00 **Minors:** Business Analytics and Mind & Behavior
Honors/Awards: Seven terms High Honor Certificate (GPA higher than 3.5), %100 Scholarship (Outstanding Achievement Scholarship)

SKILLS

Programming: Python, C++ (Advanced), SQL, JavaScript, Dart, Verilog, Laravel, Flutter
AI / ML: PyTorch, scikit-learn, Hugging Face Transformers, gensim, FAISS, RAG, LLM agents (ReAct), NumPy
Web & Tools: FastAPI, Next.js, React, Node.js, HTML/CSS, PostgreSQL, Docker, Celery/Redis, Git, MongoDB, Bootstrap
Languages: Turkish (Native), English (Advanced), German (Beginner)

WORK EXPERIENCE

Nov 2025 – Present **HEPSIBURADA Istanbul, Turkey** – ***Data Science Intern***

- Improved response diversity of LLM-based chatbots across multi-turn conversations while preserving factual accuracy.
- Reduced hallucinations and topic drift in long-context conversations, improving dialogue coherence and stability.
- Corrected position bias in query autocomplete ranking with Inverse Propensity Weighting and Doubly Robust learning, producing a fairer and more relevant suggestion ranking.
- Contributed to an event migration project, restructuring existing events into a new event schema.

Jun – Aug 2025 **Albasoft Information Technologies Ankara, Turkey** – ***Full-Stack Developer Intern***

- Developed a Hospital Information Management System with Laravel (MVC), enhancing UI responsiveness with Bootstrap and adding interactive features with jQuery.

Feb – Jun 2024 **Sabanci University Istanbul, Turkey** – ***Learning Assistant: Data Structures, Calculus-II***

- Supported students in Data Structures and Calculus-II through problem-solving sessions, clear explanations, and office hours.

Jan - Feb 2024 **TUBITAK BILGEM Istanbul, Turkey** – ***Intern***

- Worked on radio signal application systems, solving data structures and algorithm tasks in C++ and gaining hands-on hardware design experience with Verilog.

PROJECTS

Feb 2026 – Present **Know First, Then Act: LLM Agent with an Epistemic Critic**

- Built an LLM agent in AI2-THOR that splits household tasks into subgoals and uses an Epistemic Critic to check, before every step, whether the knowledge it needs is in memory, searching for missing objects instead of failing first; in a pilot study it solved 18/18 episodes and caught 98% of real knowledge gaps with zero false alarms.

May – Jun 2026 **SuSchedule-r, LLM-Based Course-Planning Agent**

- Built a GPT-4o ReAct agent for Sabanci University course planning with a 22-tool typed toolbox and RAG retrieval (BAAI bi-encoder + FAISS) over 817 courses, reaching 0.977 Hit@5 retrieval and 91.7% end-to-end answer grounding.

Mar – Apr 2026 **Content Viral, YouTube-to-TikTok Automation Pipeline**

- Built a full-stack pipeline (Next.js 14, FastAPI, Celery/Redis) turning trending YouTube videos into TikTok-ready clips via yt-dlp/ffmpeg and Whisper/GPT for AI segmentation, captions & hashtags.

Feb – Mar 2026 **SubsAI, Subscription Tracking Platform**.

- Built a Dockerized full-stack tracker (FastAPI, PostgreSQL, Next.js, Celery/Redis, JWT) with a GPT-4o-mini structured-output pipeline extracting subscriptions from PDF/OCR statements after regex-masking IBANs and card numbers, plus human-in-the-loop review and payment reminders.

Jan – Feb 2026 **MovieRAG, Retrieval-Augmented Generation Pipeline**

- Built a two-stage RAG pipeline over 41K movies (BGE + FAISS top-20, cross-encoder re-ranking to top-5, Qwen2.5-3B-Instruct 4-bit); re-ranking raised Recall@5 from 0.70 to 0.85 and MRR from 0.59 to 0.72, and a strict prompt cut hallucinations on unanswerable queries to 0%.

Nov – Dec 2025 **Song2Vec, Music Embedding & Recommendation System**.

- Trained skip-gram Word2Vec embeddings on 48K playlists (440K tracks) across 22 configurations (best HR@10: 12.6%), adding an artist-vector cold-start fallback and diversity re-ranking.